

Assignment 4

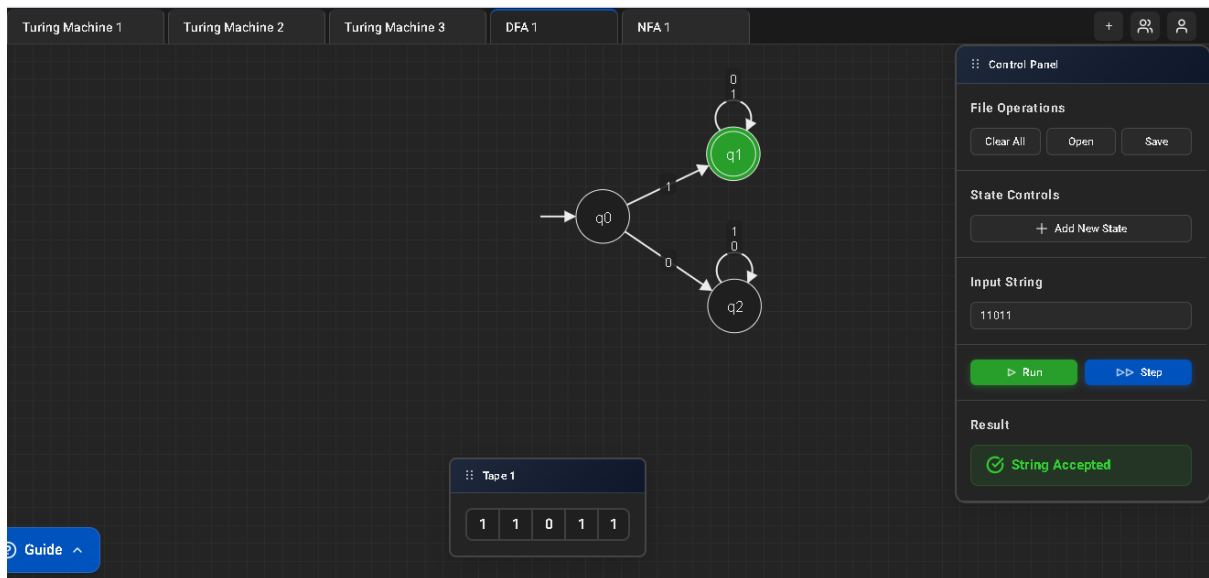
Name: Muhammad Bilal

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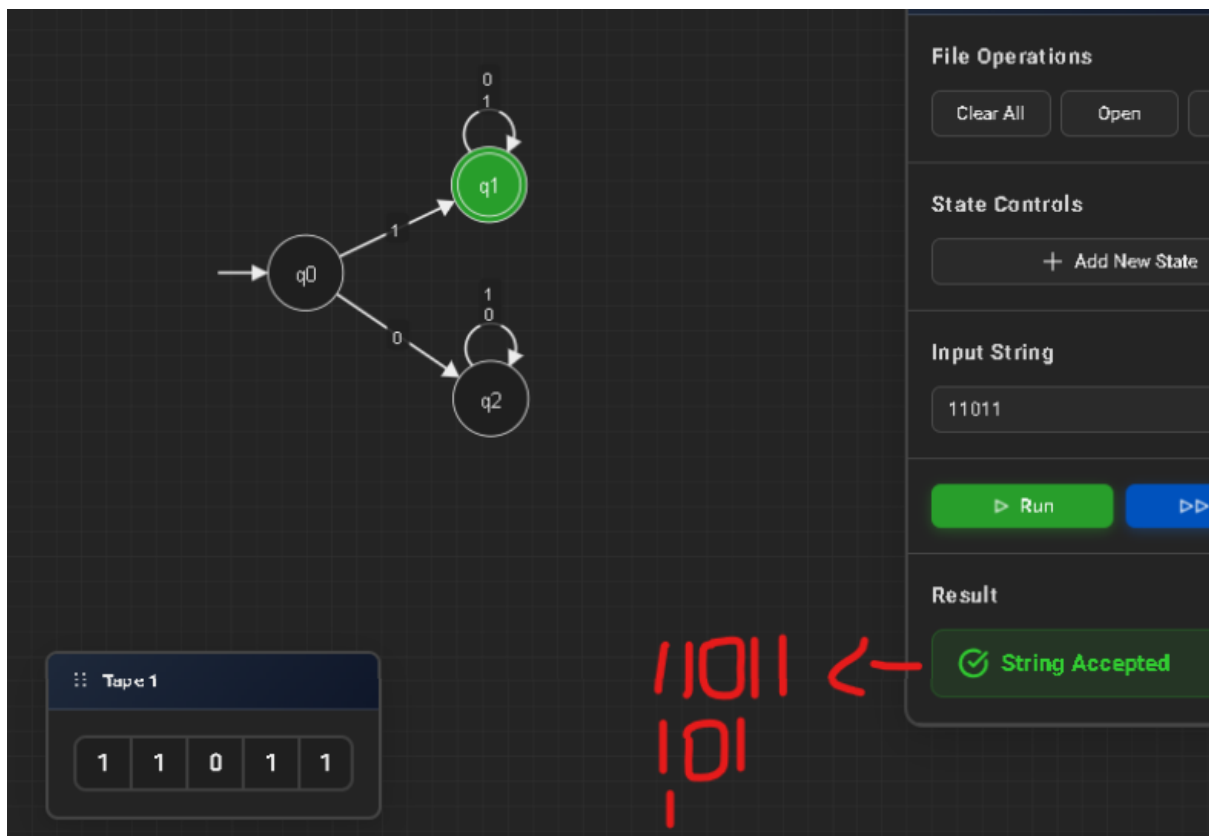
Question 1:

RE: $^1(0|1)^*$

DFA

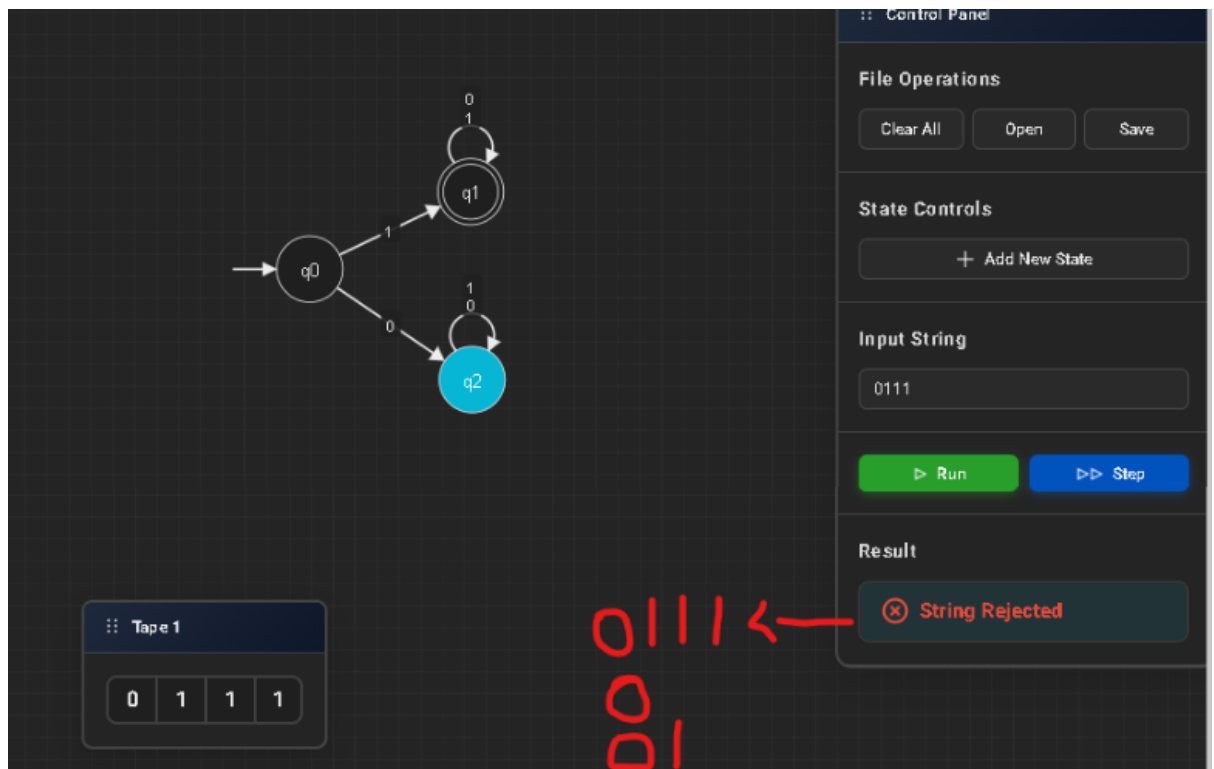


Accepted strings eg:



one more example is 100

1. Rejected string eg:

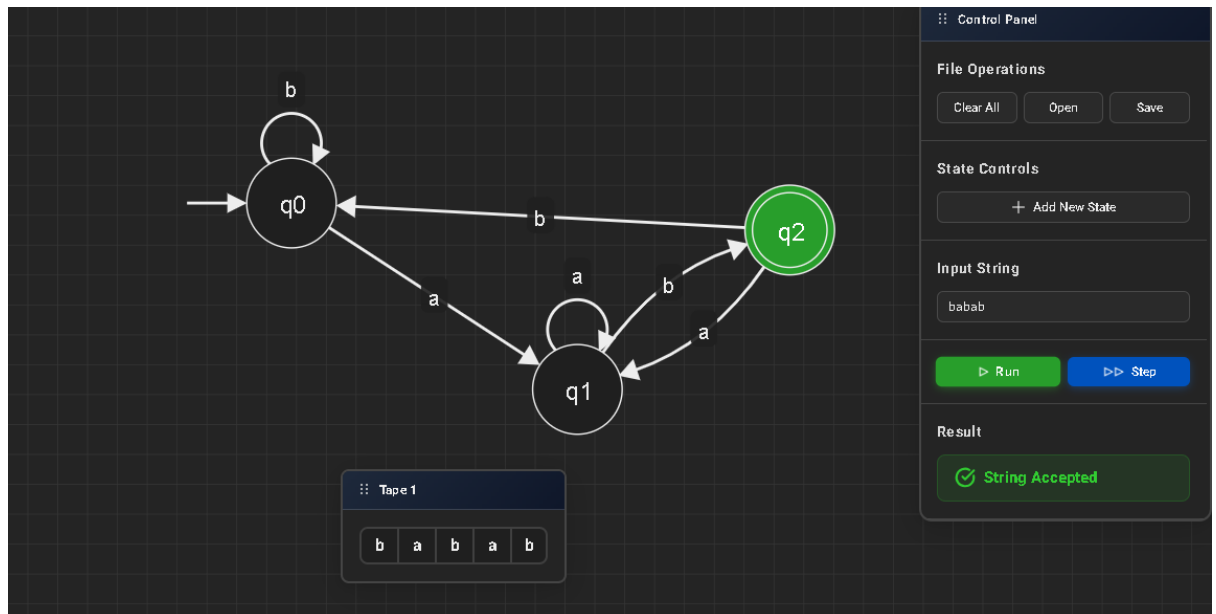


one more example is 0101

Question 2:

RE: $\wedge(a|b)^*ab\$$

DFA:



accepted strings:

1. babab
2. aaab
3. ab
4. bab

Rejected string eg:

a

ba

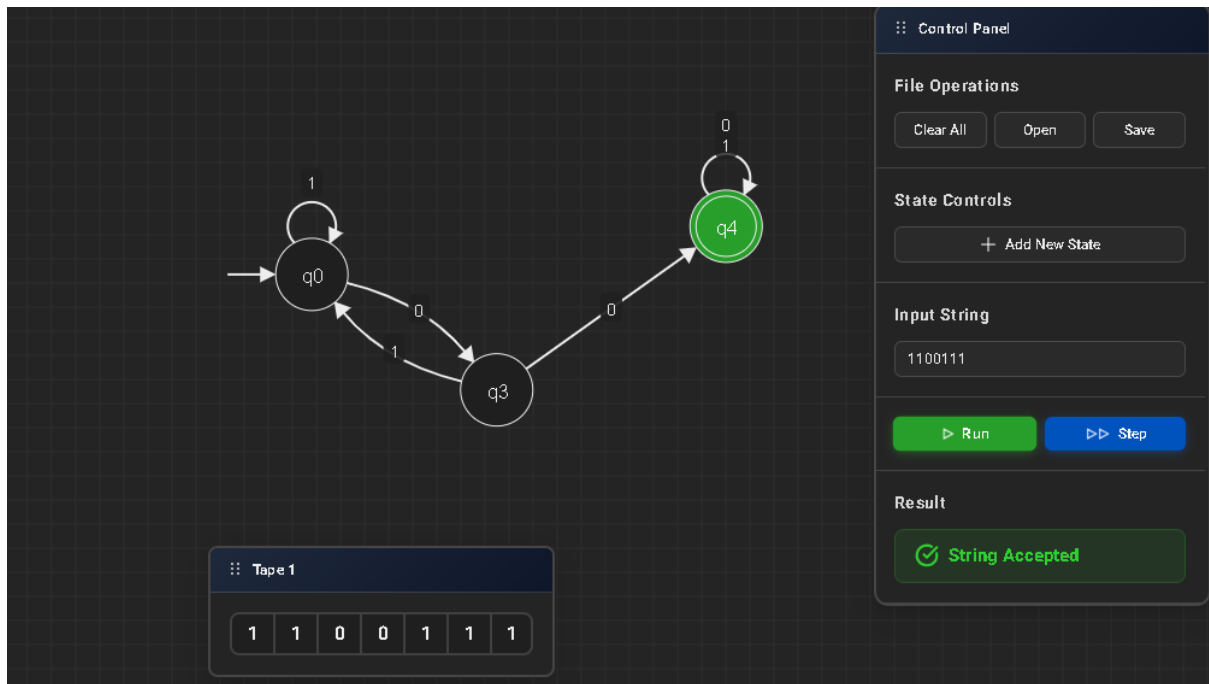
aba

abb

Question 3:

RE: $\wedge(0|1)^*00(0|1)^*$

DFA:



Accepted strings eg:

1100111, 1001, 00, 000

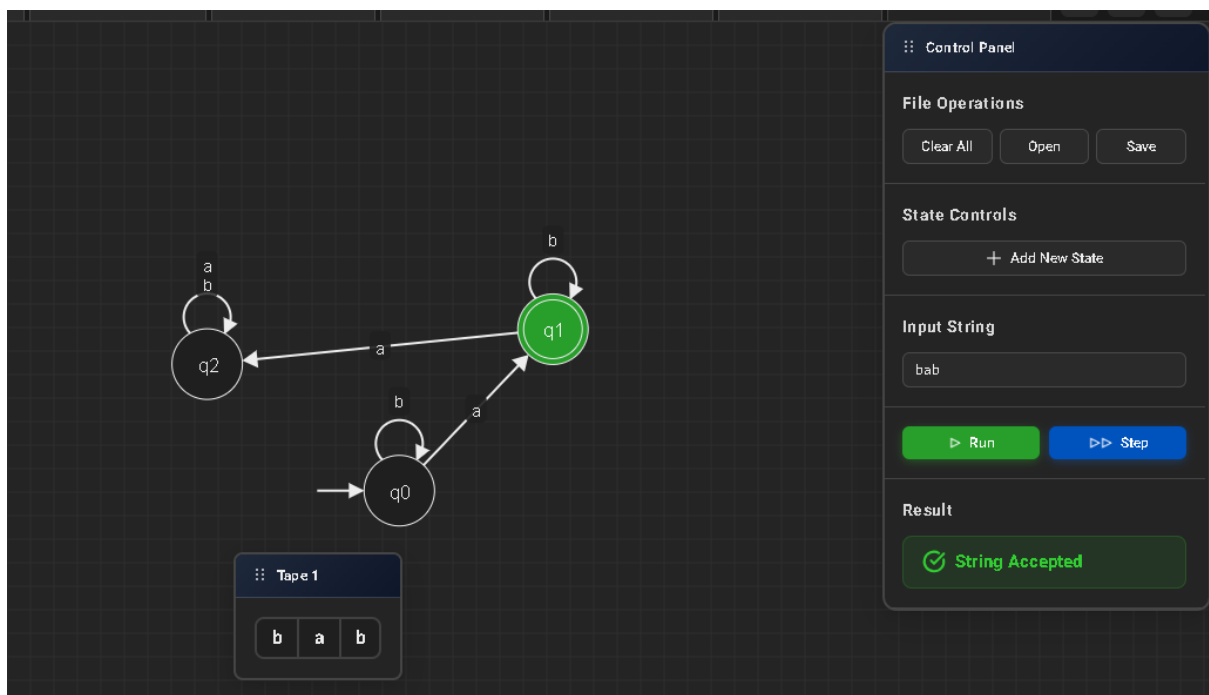
Rejected string eg:

01, 101, 1, 111

Question 4:

RE: $^ab^*ab^*$

DFA:

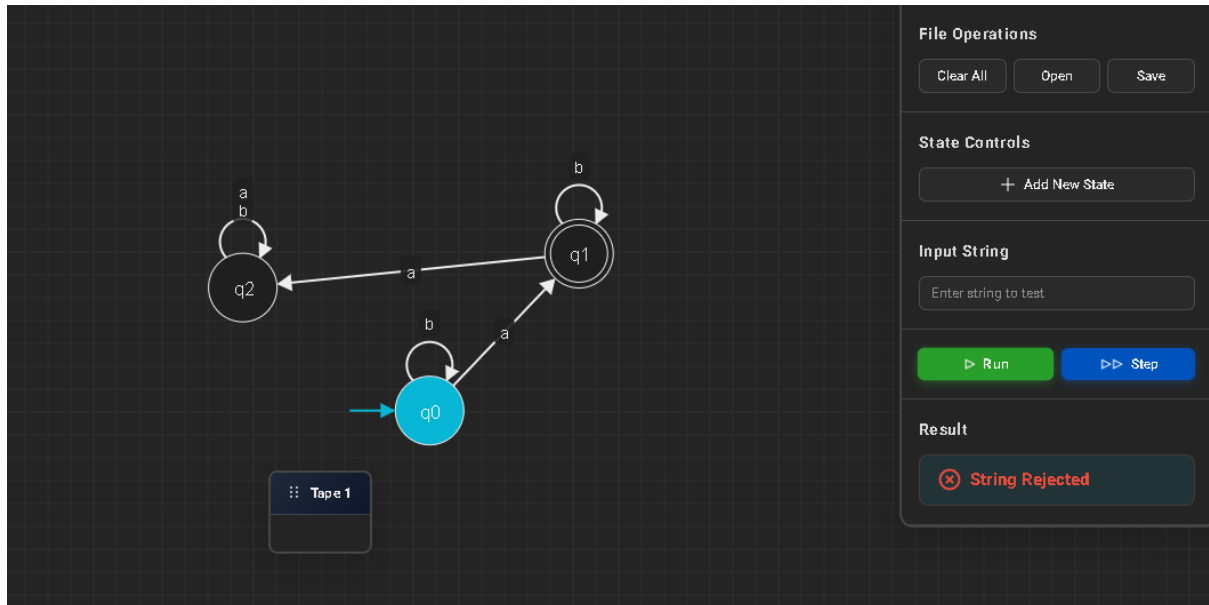


Accepted strings eg:

a, bab, bba, abbb

Rejected string eg:

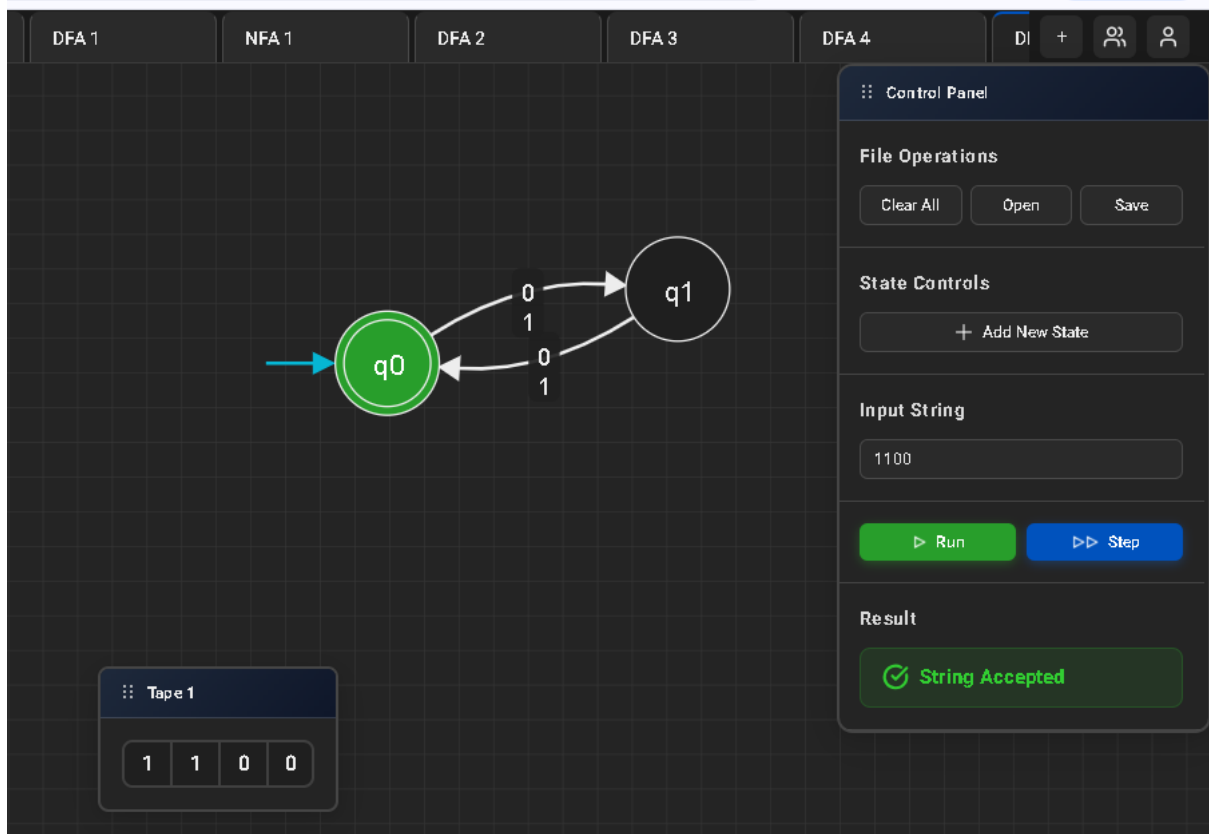
aa, aab, b, empty



Question 5:

RE: $^((0|1)(0|1))^*$

DFA:

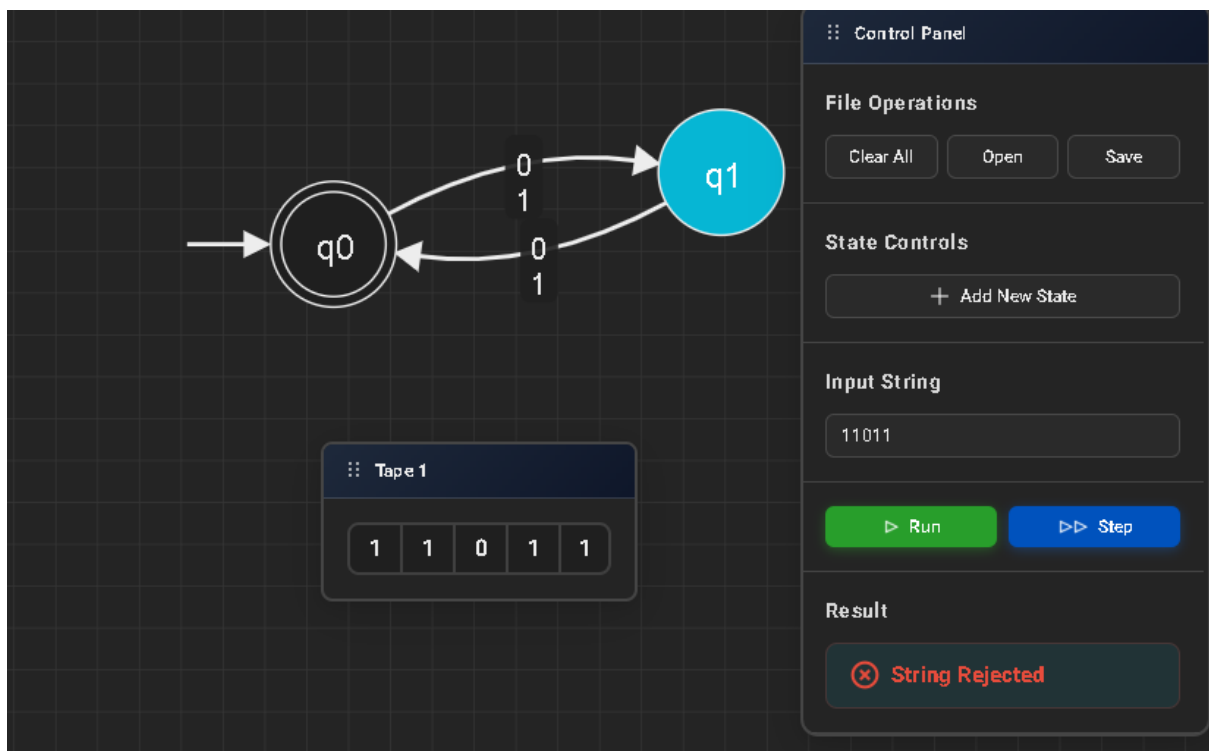


Accepted strings eg:

1100, empty, 01, 10

Rejected string eg:

0, 1, 010, 11011

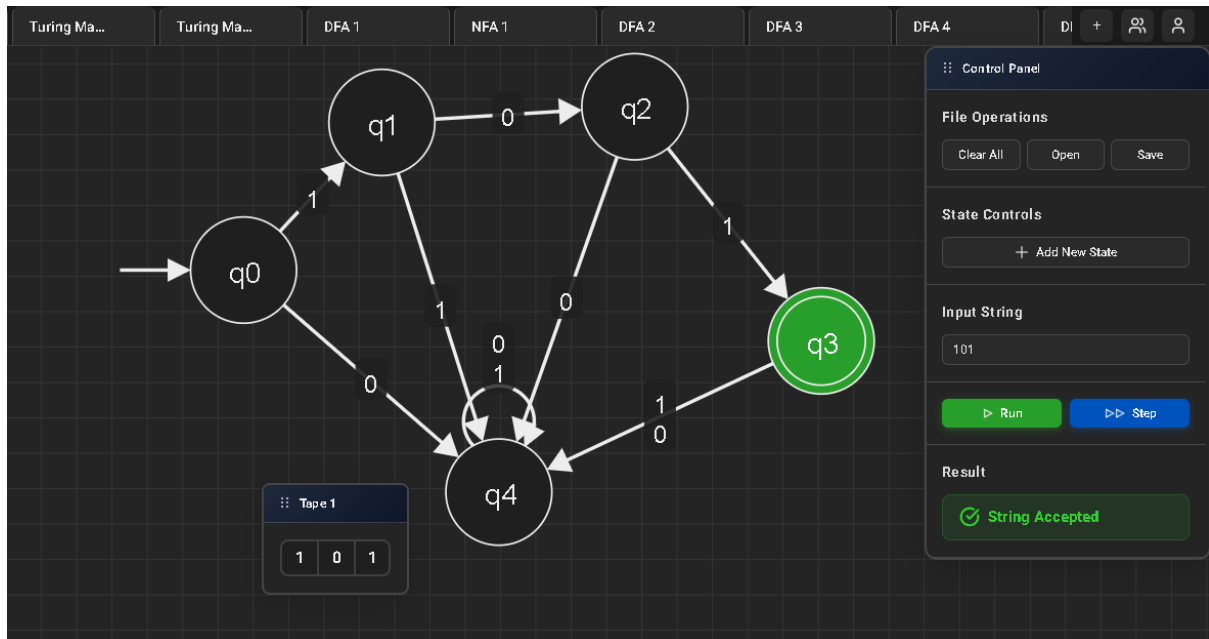


Question 6:

Exact same

RE: 101

DFA:



Accepted string:

101

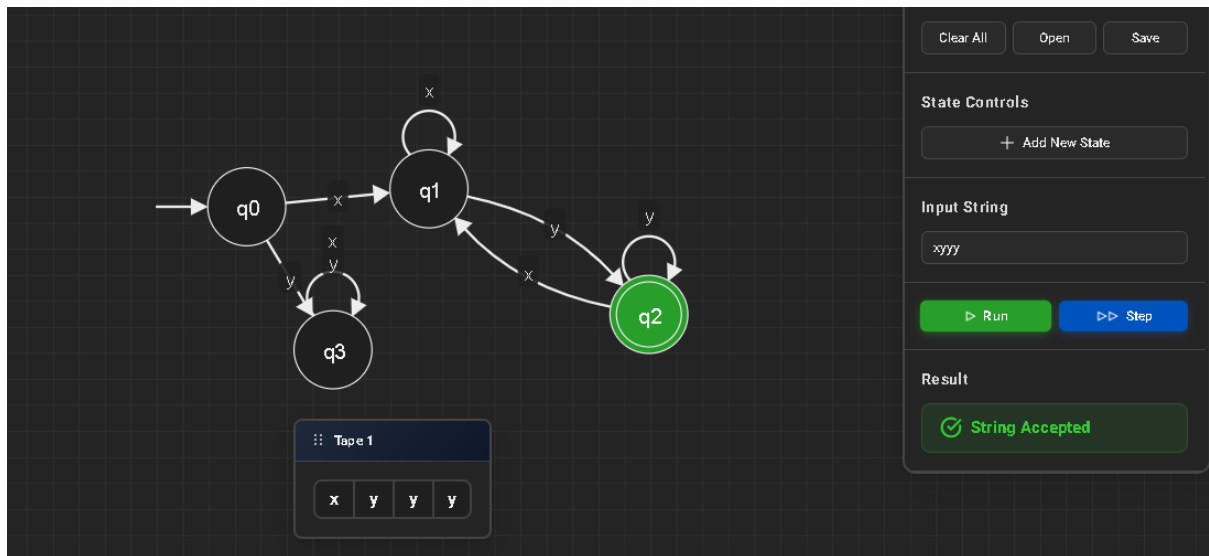
Rejected string eg:

10, 1011 any string except 101 is rejected

Question 7:

RE: $^x(x|y)^*y\$$

DFA:



Accepted strings eg:

xyxy, xyyy

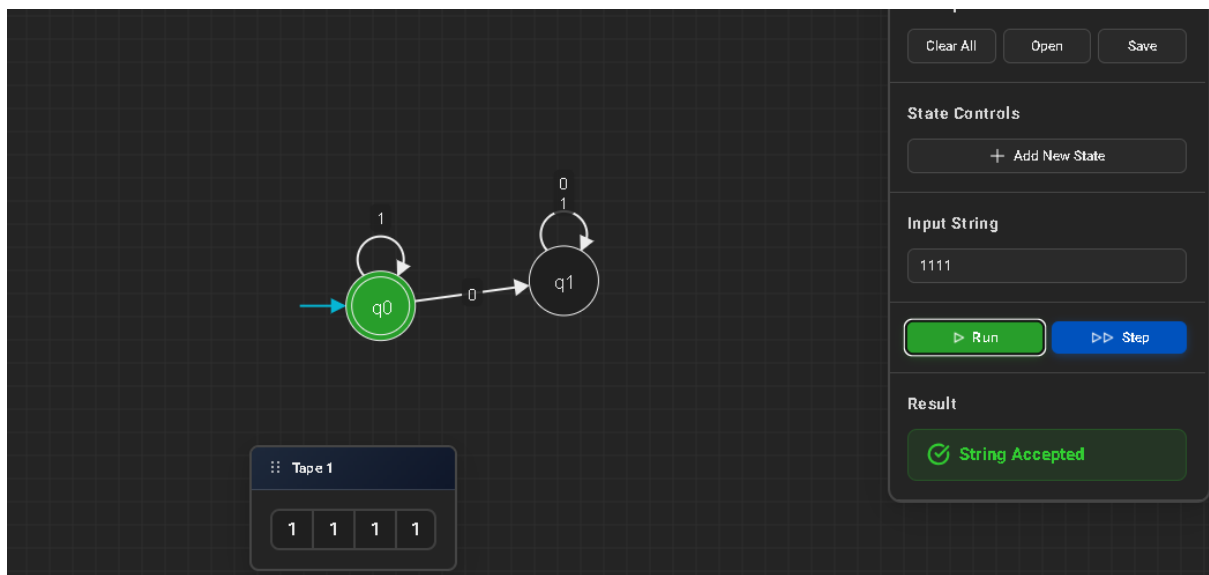
Rejected string eg:

yyxy, yx, xx, y

Question 8:

RE: $^11^*$

DFA:



Accepted strings eg:

1,111,111

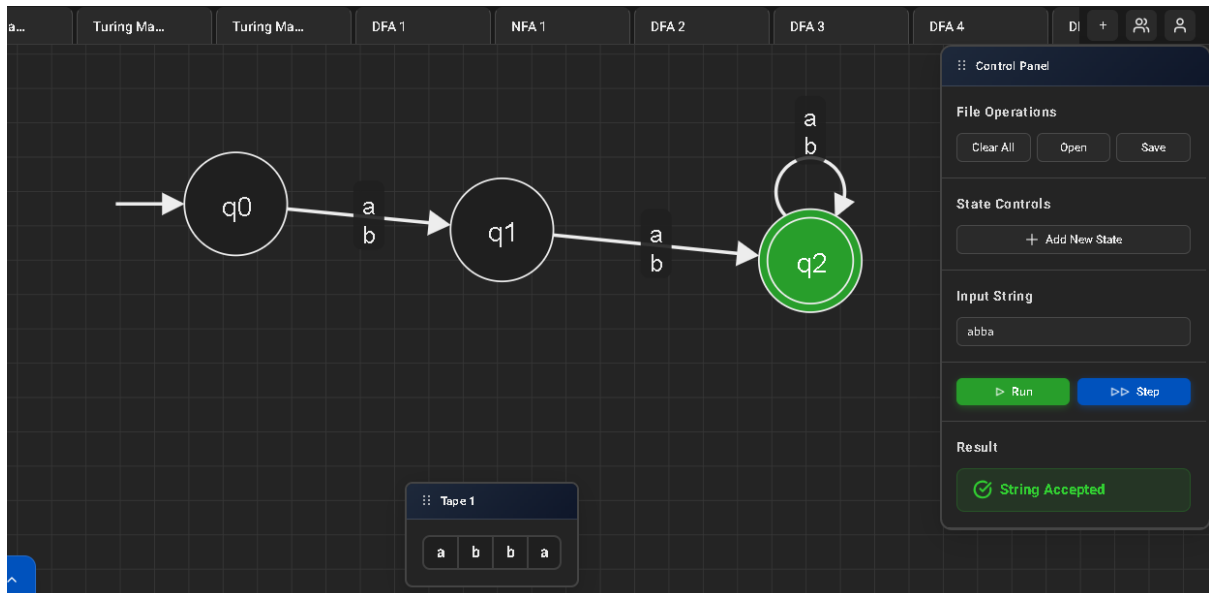
Rejected string eg:

110, 0110, 10

Question 9:

RE: $^a(a|b)(a|b)(a|b)^*$

DFA:



Accepted strings eg:

ab, abba

Rejected string eg:

a, b, empty string

Question 10:

RE: $^0[01]^*\$|^[01]^*1\$$

DFA:

